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Exercise 1 SQL fundamentals (SELECT & Filtering)

Selecting & Sorting.

1. Syntax: SELECT *
FROM employees;

Table employees

ID	First_name	Last_name	Department	Salary	Hire_date	City
1	Alice	Green	IT	70000	2020-01-10	JHB
2	Brian	Lee	HR	45000	2019-03-22	CT
3	Cathy	Zulu	Finance	65000	2018-07-18	DUR
4	David	Mokuwena	Marketing	50000	2021-11-05	PTA
5	Eva	Naidoo	IT	72000	2021-09-30	JHB

2. Syntax: SELECT DISTINCT department,
FROM employees;

Table employees

Department
IT
HR
FINANCE
MARKETING.

3. Syntax: SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

Table employees

first_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. Syntax : SELECT id,
 first_name,
 last_name, salary
 FROM employees
 ORDER BY salary DESC
 LIMIT 3;

Table employees.

ID	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	10000
3	Cathy	Zulu	65000

4. Filtering with WHERE

5. Syntax : SELECT id,
 first_name,
 last_name, department
 FROM employees
 WHERE department = "IT";

Table employees

ID	first_name	last_name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Syntax : SELECT id,
 first-name,
 last-name, department, salary
 FROM employees
 WHERE department = "Finance" AND salary > 60 000;

Table employees

ID	first-name	last-name	Department	Salary
3	Cathy	Zulu	Finance	65 000

7. Syntax : SELECT id,
 first-name,
 last-name,
 department
 FROM employees
 WHERE department = "HR" OR "Marketing";

Table employees

ID	first-name	last-name	department
2	Brian	LEE	HR
4	David	Mokoena	Marketing

8. Syntax : SELECT id,
 first-name
 last-name
 FROM employees
 WHERE department != "IT";

~~Table emplo~~

Table employees

ID	first_name	Last_name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Syntax : SELECT id,
first_name
last_name, department
FROM employees
WHERE department IN "IT", "HR", "FINANCE";

Table employees

ID	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Syntax : SELECT id,
first_name
last_name
department
Salary
City
FROM employees
WHERE department = "IT" AND salary > 65000
AND city "Johannesburg";

Table employee

ID	first_name	last_name	Department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg